

Suggested Solutions to Exercises 1

1. The future value after n years with annual compounding is given by the formula $FV_n = PV(1+r)^n$, where PV is the present value and r is the annual interest rate. Therefore, after n years at 10% compounded annually £1,000 will accumulate to $£1,000(1+0.1)^n$. For example:

n	£
1	1,100
2	1,210
3	1,331
4	1,464.10
5	1,610.51

2. Using the formula for annual compounding introduced in Exercise 1, we obtain $FV_n/PV = (1+r)^n$. In this example, $FV_n = 1,039.50$, $PV = 500$ and $r = 0.05$, and therefore n can be obtained solving

$$\frac{1,039.50}{500} = (1+0.05)^n.$$

Taking logs on each side we obtain:

$$\log \frac{1,039.50}{500} = n \log(1+0.05),$$

which gives:

$$n = \frac{\log \frac{1,039.50}{500}}{\log(1+0.05)} = 15.$$

In conclusion, it will take 15 years for £500 to grow to £1,039.50 if invested at 5% compounded annually.

3. We want to borrow some money, and we are offered two different interest rates. We have to compare the two rates and then choose the lower one, since this would generate a lower total repayment.

We know that in the case of $r = 13\%$ compounded annually, the annual interest is 13%. But what about the case of non-annual compounding? We know that if we compound m times a year, after one year the value increases to:

$$FV_1 = PV \left(1 + \frac{r}{m}\right)^m.$$

This means that the interest for one year is $(1 + r/m)^m - 1$. In the case of a rate $r = 12\%$ and monthly compounding ($m = 12$) we obtain an interest for 1 year equal to:

$$\left(1 + \frac{0.12}{12}\right)^{12} - 1 = (1.01)^{12} - 1 = 1.12683 - 1 = 0.12683,$$

i.e. 12.683%, which is less than 13%.

Therefore we conclude that the best deal is to borrow money at a rate of 12% compounded monthly.

4. To compute the present value in the case of continuous compounding we have to use the formula $PV = FV_n \cdot e^{-r \times n}$. In this example we have $n = 8$, $r = 0.03$ and $FV_8 = 1,000$. Therefore the present value can be computed as:

$$PV = 1,000 \cdot e^{-0.03 \times 8} = 786.63.$$

We conclude that the present value of £1,000 to be received after 8 years if we discounted back to present at 3% per annum with continuous compounding is £786.63.

5. A coupon of 10% paid semi-annually means that we receive 10% of the principal every year, in two payments of 5% each, in the 6th and 12th month. In our case, the principal is £1,000, and we have a 5-year bond. It follows that we expect ten payments of £50 after 6, 12, 18, 24, ..., 48, 54, 60 months, plus a final payment of £1,000 on the 60th month. To compute the present value of the bond, we need to discount back to present each payment at a rate of r with continuous compounding, i.e. using a discount factor of $e^{-n \times r}$, where n is the time in years.

If the discount rate is 8%, then:

$$\begin{aligned}
 PV &= \sum_{i=1}^{10} 50 \cdot e^{-\frac{6i}{12} \times 0.08} + 1,000 \cdot e^{-5 \times 0.08} \\
 &= 1,074.23.
 \end{aligned}$$

When the discount rate is 8.25% then:

$$\begin{aligned}
 PV &= \sum_{i=1}^{10} 50 \cdot e^{-\frac{6i}{12} \times 0.0825} + 1,000 \cdot e^{-5 \times 0.0825} \\
 &= 1,063.31.
 \end{aligned}$$

(See values in the table.)

We conclude that when the rates go up, the value of bonds goes down.

Months	Years	CF	DCF at 8%	DCF at 8.25%
6	0.5	50	48.03	47.98
12	1	50	46.16	46.04
18	1.5	50	44.34	44.18
24	2	50	42.61	42.39
30	2.5	50	40.94	40.68
36	3	50	39.33	39.04
42	3.5	50	37.79	37.46
48	4	50	36.31	35.96
54	4.5	50	34.88	34.49
60	5	1050	703.84	695.09
		Total	1074.23	1063.31